TED University | Department of Computer Engineering

Project Proposal

Design and Development of a Smart Cane Attachment with Obstacle Detection and User Alerting for Visually Impaired Individuals

CMPE 491 Senior Design Project I | Group 18 | 16 October 2026

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Project website: https://mertsaglm.github.io/tedu-smart-cane/

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Project description

We propose a detachable electronic attachment for existing white canes that will help visually impaired individuals notice obstacles at chest and head height. During an initial visit to an association for people with visual impairments, members of our team heard that upper-body obstacles were a recurring difficulty. This motivates our focus on protruding signs, tree branches and similar objects that may lie beyond the ground-level contact of a conventional cane. The proposed device will estimate obstacle proximity and provide understandable haptic and/or audible alerts, with feedback becoming more noticeable as an obstacle approaches. We aim to develop a compact, affordable attachment that preserves the familiar handling and ground-contact function of the white cane.

Our development will investigate ultrasonic and/or time-of-flight distance sensing, a microcontroller, and inertial measurements to account for cane orientation and sweeping motion. Sensor selection, mounting geometry and the feedback method will be refined through experiments and user feedback. The engineering focus is to investigate whether motion-aware filtering can reduce unnecessary alerts while retaining timely obstacle warnings. We will assess detection performance, missed obstacles, false alerts, response time, added weight, battery use and attachment stability in defined indoor and outdoor scenarios. The initial prototype materials budget is approximately TRY 10,000, subject to component selection and quotations. Object-type recognition and stair or drop-off detection will be explored as possible extensions rather than guaranteed capabilities of the initial prototype. CMPE 491 will establish the requirements, analysis, high-level design and an initial demonstration; implementation and evaluation will continue in CMPE 492. The intended outcome is a documented engineering prototype that complements established cane use.

Initial literature review

Existing electronic travel aids establish several relevant baselines. WeWALK combines upper-body obstacle detection with navigation and voice assistance [1]. GoSense Rango is a detachable white-cane module that conveys obstacle position through spatial audio [2]. SmartCane, developed at IIT Delhi with its partners, uses ultrasonic sensing and distance-dependent vibration patterns [3]. These products show that obstacle sensing and a cane attachment alone are established concepts. Our contribution will therefore be evaluated through the practicality of the mounting, cost-conscious design and measured behaviour during cane movement.

More complex systems illustrate the trade-offs involved in extending the scope. Glidance proposes physical steering through its wheeled Glide mobility aid [4]. Stanford researchers demonstrated an augmented cane using robotic sensing and steering [5]. In China, Mai et al. combined 2D LiDAR, an RGB-D camera and object recognition in a wheeled smart cane [6]. The latter approaches provide useful references for navigation and richer perception, but introduce hardware, processing and mechanical requirements beyond a small detachable warning device.

These findings support prioritising upper-body obstacle awareness and comparing a simple distance-threshold baseline with an approach that considers cane motion. Subsequent analysis will examine sensing limitations, feedback usability and controlled evaluation methods. Manufacturer descriptions establish advertised capabilities; they will not be treated as independent evidence of detection accuracy or as proof of the performance of our prototype.

References

[1] WeWALK. Smart Cane 2 product overview. Accessed 16 October 2026.

[2] GoSense. Rango product sheet. Accessed 16 October 2026.

[3] Saksham Trust. SmartCane overview. Accessed 16 October 2026.

[4] Glidance. Glide product overview. Accessed 16 October 2026.

[5] Stanford University School of Engineering. A team of researchers build a $400 self-navigating smart cane, 2021. Accessed 16 October 2026.

[6] C. Mai et al.. A Smart Cane Based on 2D LiDAR and RGB-D Camera Sensor-Realizing Navigation and Obstacle Recognition. Sensors, 24(3), 870, 2024. DOI: 10.3390/s24030870. Accessed 16 October 2026.